



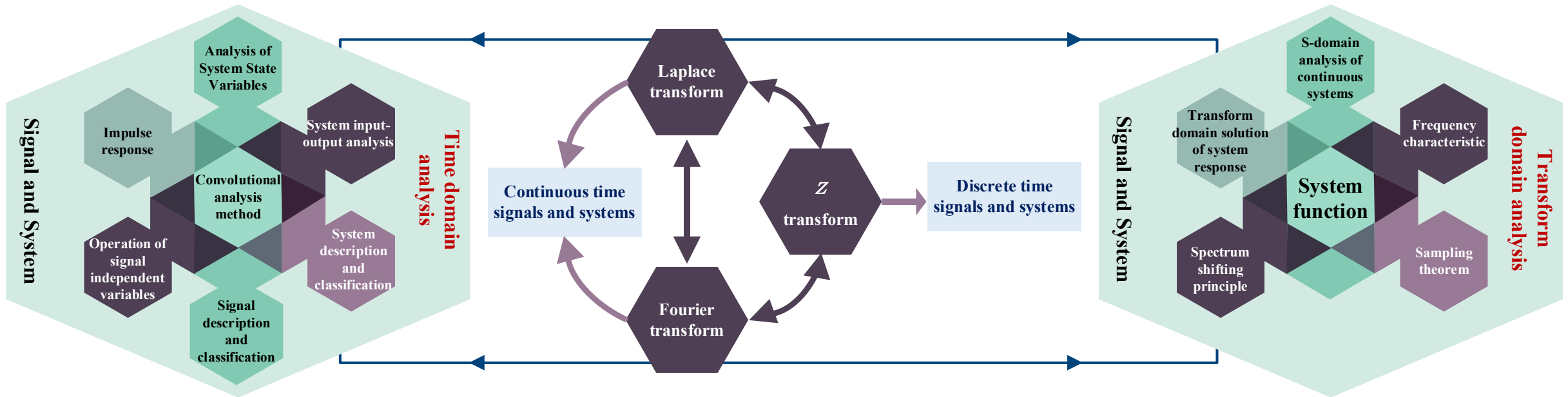
香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

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1. Mind Map



- ❑ It serves as the foundation for subsequent studies in specialized courses such as 'Principles of Automatic Control', 'Digital Signal Processing', and 'Artificial Intelligence'
- ❑ Basic course in engineering-related disciplines, relying on courses such as 'Advanced Mathematics', 'Linear Algebra', and 'Fundamentals of Circuit Analysis'
- ❑ Pre-requisite: MATH2011 or MATH2023 or MATH2111 or MATH2350 or MATH2351 or MAT2352
- ❑ Description methods and processing techniques of signals and systems in the time domain and transform domain

2. Learning Goals

Intended Learning Outcomes (ILOs)

CO1 - Describe Basic Continuous Time and Discrete Time signals and different ways to make use of and manipulate them

CO2 - List the Properties of LTI systems and to determine the output of an LTI system using the impulse response and the convolution sum/integral, and the frequency response and transform

CO3 - Correctly apply the appropriate transform (FS, DTFS, FT, DTFT) to produce a Frequency domain representation for continuous-time/discrete-time and periodic/apperiodic signals, and relate basic operations in the time and frequency domains

CO4 - State and prove the sampling theorem

CO5 - Analyze differential and difference equations as causal LTI systems and to realize them in different block diagram forms

CO6 - Apply theories learnt to the analysis of communication systems including Amplitude Modulation and Frequency Division Multiplexing, the communication channel, mechanical systems, and new problems

CO7 - Use the Software Tools MATLAB to manipulate, process, analyze and plot signals

1. Overview

Assessments:

Homework	18%
Laboratory exercises	12%
Online Quiz	5%
Mid-term Examination	25%
Final Examination	40%

❑ Online Quizzes (Total: **5**)

- **Three** quizzes (refer to course schedule on canvas)
- **Two** quizzes **in-class**: Before the midterm exam and the final exam

❑ Midterm Exam: A **1.5-hour exam** consisting of 4-5 problem sets (multiple sub-questions)

❑ Final Exam: A **3-hour exam** consisting of 5-6 problem sets (multiple sub-questions)



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Thank you for listening !

